BrandPulse AI

Sentiment Analysis on Twitter Data (NLP)

Project Report

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Domain
Artificial Intelligence

Persevex Internship
August 2026

Abstract

Social media platforms such as Twitter (X) generate a large amount of customer feedback every day, making it difficult for marketing teams to manually track how people feel about a brand. This project presents **BrandPulse AI**, an end-to-end Natural Language Processing system that automatically classifies tweets as Positive, Negative, or Neutral.

The project compares two approaches to sentiment analysis. The first uses **TF-IDF with Logistic Regression**, while the second uses a **Bidirectional LSTM** for deep learning. Both models were trained and evaluated using the **Twitter US Airline Sentiment dataset**, which contains 14,640 tweets. The complete workflow includes text preprocessing, exploratory data analysis, feature engineering, model training, evaluation, error analysis, and deployment.

To make the system easier to use, the trained models were integrated into a **Streamlit dashboard**. The dashboard supports single-tweet predictions, batch analysis using CSV files, a simulated live-tweet stream, a 24-hour sentiment trend, and a comparison of the two models.

The Bidirectional LSTM achieved the best test accuracy at **80.0%**, compared with **77.4%** for Logistic Regression. However, this improvement came with higher computational cost, including roughly 50 times longer training time and a model size around 35 times larger. Logistic Regression also remains easier to interpret. For both models, **Neutral** tweets were the most difficult to classify, with an F1-score of around 0.60. Based on these results, the project also provides practical recommendations for choosing between the two approaches depending on accuracy, speed, and interpretability requirements.

Introduction

Social media platforms such as Twitter (X) generate a huge amount of customer feedback every day. This makes it difficult for marketing and customer-support teams to manually track what customers are saying about a brand. Sentiment analysis can help by automatically identifying whether a tweet is Positive, Negative, or Neutral.

This project was developed as part of an NLP engineering internship, with a focus on building and comparing two different approaches to sentiment analysis. The first approach uses **TF-IDF with Logistic Regression**, which is fast and easy to interpret. The second uses a **Bidirectional LSTM**, which can understand word order and context better but requires more training time and resources.

The main aim of the project is not only to build a working sentiment classifier, but also to understand the practical differences between these two approaches and determine when each model is more suitable.

Problem Statement

BrandPulse AI is an analytics case study focused on understanding customer sentiment about airline services on Twitter. With a large number of tweets being posted every day, marketing and customer-support teams cannot manually monitor all customer feedback or quickly identify complaints that need attention. This can delay responses to service issues while positive and neutral conversations add to the overall volume of data.

Objectives

- **Build a tweet preprocessing pipeline** to handle URLs, mentions, hashtags, lemmatization, and stopword removal while preserving important negations.
- **Classify tweets** into Positive, Negative, or Neutral sentiment using supervised machine learning.
- **Compare two approaches**, TF-IDF with Logistic Regression and a Bidirectional LSTM.
- **Evaluate both models** using Accuracy, Precision, Recall, F1-score, and confusion matrices.
- **Analyze model errors** to identify the types of tweets that are difficult to classify.
- **Interpret the models** by identifying the words that have the strongest influence on each sentiment class.
- **Deploy the models** through an interactive Streamlit dashboard for single and batch predictions.
- **Derive practical recommendations** based on model performance, speed, interpretability, and limitations.

1. Dataset Description

The project uses the Twitter US Airline Sentiment dataset, which contains real tweets directed at six major US airlines, originally collected and labeled by Crowdflower in February 2015.

Attribute	Value
Dataset	Twitter US Airline Sentiment
Total Records	14,640 tweets (14,619 after cleaning)
Airlines covered	6 major US airlines
Target Variable	Airline sentiment (negative / neutral / positive)
Source	Crowdflower (Feb 2015), mirrored on GitHub / Kaggle

Important Fields Used

- **text -** The original tweet content used for sentiment analysis.
- **airline_sentiment -** The target label: Negative, Neutral, or Positive.
- **airline -** The airline mentioned or targeted in the tweet.
- **negativereason -** The reported reason for a negative tweet, when available.

The dataset is **highly imbalanced**, with negative tweets making up the majority of the data. This was taken into account during model training and evaluation.

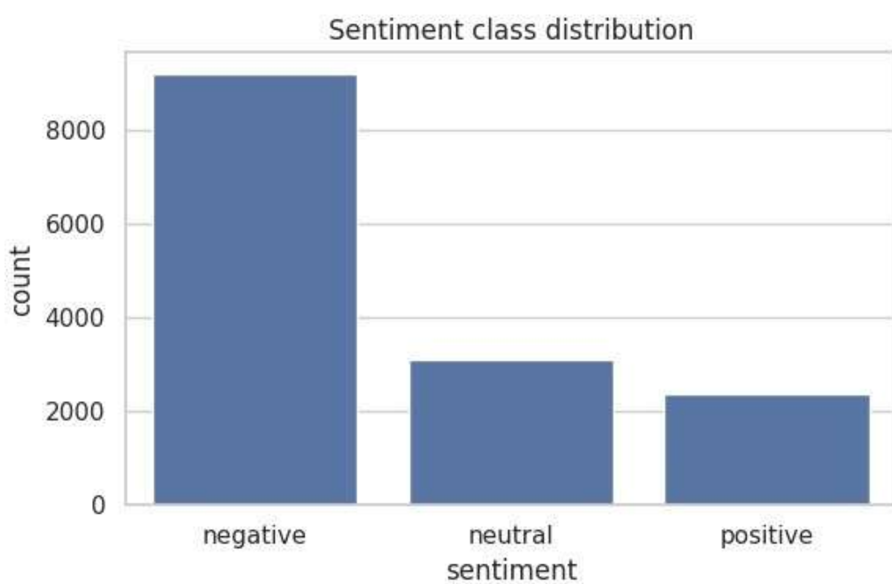

Figure 1. Sentiment class distribution negative tweets dominate the dataset.

- Negative: 9,178 tweets (62.7%)
- Neutral: 3,099 tweets (21.2%)
- Positive: 2,363 tweets (16.1%)

Data Preprocessing

Tweets contain informal language, hashtags, mentions, links, and other noise, so they need to be cleaned before training. The same preprocessing pipeline is used during both training and prediction to keep the model inputs consistent.

The pipeline includes:

- **Lowercasing** all text for normalization.
- **Removing URLs and @mentions** using regular expressions.
- **Converting hashtags to words**, for example, #delayed → delayed, instead of removing them completely.
- **Removing HTML entities and unnecessary characters** such as punctuation, numbers, and emojis.
- **Removing stopwords while keeping negations** such as *not, no, never,* and *n't*, since "not good" should remain different from "good".
- **Lemmatizing words** using NLTK WordNet Lemmatizer, for example, running → run.
- **Removing empty records** after cleaning.

The cleaned data was divided using an **80:20 stratified train-test split**, with **11,695 tweets for training and 2,924 for testing**. The same split was used for both models to ensure a fair comparison.

Exploratory Data Analysis (EDA)

Because negative tweets make up **62.7% of the dataset**, the classes are highly imbalanced, so accuracy alone would not give a complete picture of model performance. Therefore, Macro F1 and other class-level metrics were also considered during evaluation. To reduce the effect of this imbalance during training, the Logistic Regression model was trained using class_weight='balanced'.

Word-frequency analysis also showed clear differences between the classes, with words such as **"delayed", "cancelled", "hour", "bag", and "hold"** appearing frequently in negative tweets, while **"thanks", "great", "awesome", and "love"** were common in positive tweets. These patterns were later examined in more detail during model interpretability.

Feature Engineering

Classical NLP: TF-IDF

For the classical approach, the cleaned tweets are converted into numerical features using TF-IDF (Term Frequency-Inverse Document Frequency). Both unigrams and bigrams are used to capture individual words as well as short phrases. A minimum document frequency of 2 and sublinear term-frequency scaling are applied to give more importance to words and phrases that are useful for distinguishing between sentiment classes.

Deep Learning: Tokenization and Padding

For the deep learning approach, the cleaned tweets are **tokenized using a 20,000-word vocabulary** and padded or truncated to **50 tokens**. Unlike TF-IDF, this preserves the order of words in each tweet, allowing the LSTM model to learn contextual patterns. This forms the main comparison between the two approaches: **TF-IDF treats text mainly as a collection of words, while LSTM processes it as a sequence**.

Machine Learning Models

TF-IDF + Logistic Regression (Classical Baseline)

The **Logistic Regression** model uses the TF-IDF features with class_weight='balanced' and max_iter=1000. It is fast, computationally efficient, and easier to interpret because its feature coefficients show which words contribute most toward each sentiment class.

Bidirectional LSTM (Deep Learning)

The **Bidirectional LSTM** model was built using TensorFlow/Keras. It consists of a **128-dimensional Embedding layer**, a **64-unit Bidirectional LSTM**, a **Dense layer with 64 ReLU units**, **Dropout of 0.3**, and a final **3-class Softmax output layer**. The model was trained for up to 10 epochs with early stopping based on validation loss, restoring the best-performing weights.

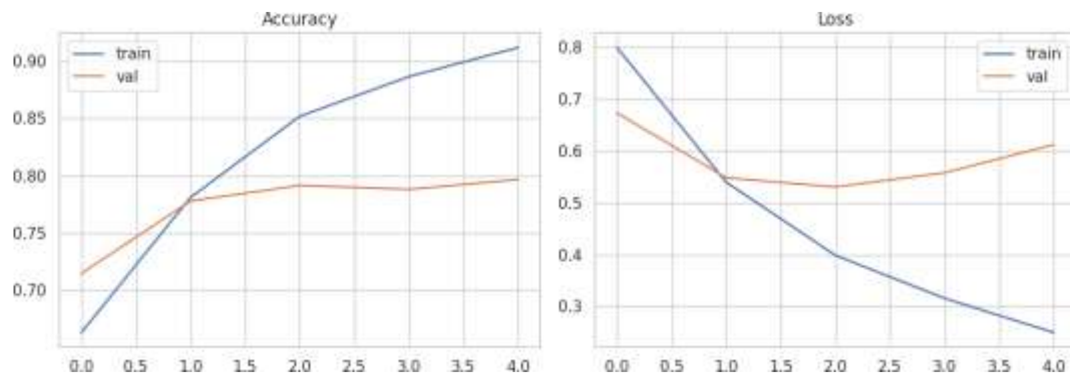

Figure 2. LSTM training curves validation loss bottoms out around epoch 3 before the model begins to overfit; early stopping restores the best weights.

Model Evaluation

Both models were evaluated on the identical held-out 2,924-tweet test set.

Model	Accuracy	Macro F1	Weighted F1
Logistic Regression	77.4%	0.72	0.78
Bidirectional LSTM	80.0%	0.74	0.80

Analysis

The **Bidirectional LSTM performed better than the Logistic Regression baseline**, achieving higher overall performance and particularly stronger recall for negative tweets, **0.89 compared with 0.83**. This improvement comes from its ability to capture word order and contextual information that TF-IDF does not consider. However, both models struggled with the **Neutral** class, which had an F1-score of around **0.60**. This shows that neutral tweets are harder to distinguish from positive and negative sentiment and remain the main limitation of both approaches.

1.1 Confusion Matrices

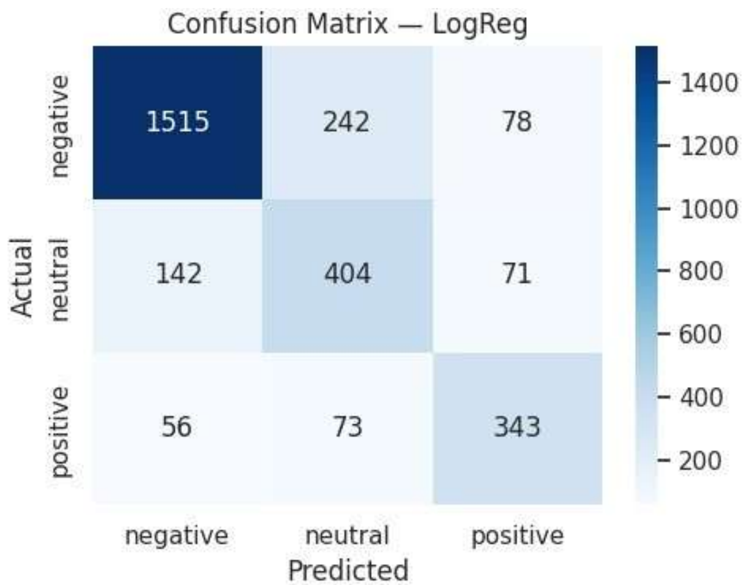

Figure 3. Confusion matrix TF-IDF + Logistic Regression.

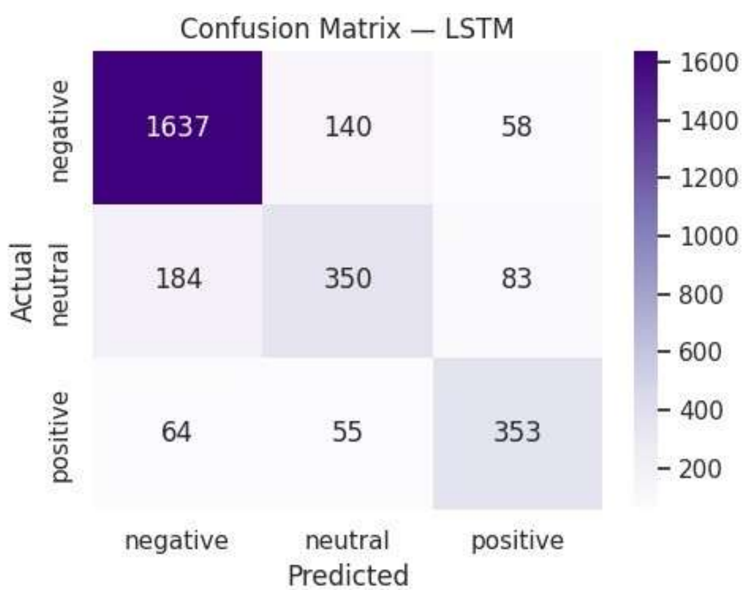

Figure 4. Confusion matrix Bidirectional LSTM.

Figures 3 and 4 show that the main source of errors for both models is confusion between **Neutral and Negative** tweets. Short, factual, or question-based tweets can contain words that are also common in complaints, making them difficult to classify correctly. In other cases, neutral tweets simply do not contain strong sentiment cues, which makes the distinction between neutral and negative sentiment more difficult.

Model Interpretability

Understanding why a model makes a prediction is important, especially when the results are used for business decisions. Since Logistic Regression assigns a coefficient to each TF-IDF feature for every sentiment class, we can examine the words with the strongest coefficients to understand which words influence the model toward **Positive, Negative, or Neutral** sentiment. This gives a simple and direct way to understand what the model has learned from the text.

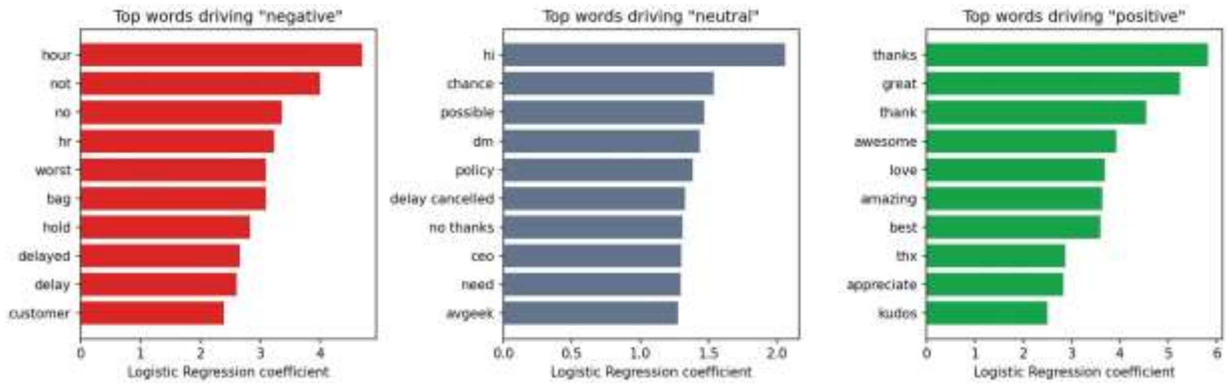

Figure 5. Top words driving each sentiment class, ranked by Logistic Regression coefficient.

The interpretability analysis shows that the Logistic Regression model has learned clear and meaningful patterns from the tweets. **Negative sentiment** is strongly associated with words such as *"hour", "worst", "bag", "hold", "delayed", "delay",* and *"luggage"*, along with negation words such as *"not"* and *"no"*. This also supports the decision to preserve negations during preprocessing.

Positive sentiment is mainly influenced by words related to appreciation and praise, such as *"thanks", "great", "thank", "awesome", "love", "amazing", "best",* and *"appreciate"*. **Neutral sentiment** is associated with more general or procedural words such as *"hi", "chance", "possible", "dm",* and *"policy"*. These words provide weaker sentiment signals, which helps explain why the Neutral class is the hardest for both models to classify.

The Bidirectional LSTM does not provide the same direct word-level coefficient interpretation because its learned representation is distributed across the embedding and recurrent layers. This highlights an important trade-off between the two models: **Logistic Regression gives up some accuracy in exchange for greater interpretability**, while the BiLSTM provides better performance but is harder to explain at the individual-word level.

Error Analysis

Both models were tested again on the held-out test set to examine the tweets they classified incorrectly. The main errors were between **Neutral and Negative** and **Neutral and Positive**, while **Negative and Positive** tweets were rarely confused with each other. This suggests that both models can identify clear sentiment reasonably well, but struggle when a tweet has weak or unclear sentiment.

For example, **"@JetBlue actually flight 1089 not 1098"** was labelled Neutral but predicted as Negative with 80.2% confidence. The tweet is simply correcting a flight number, but the model may associate the surrounding vocabulary with complaints. Similarly, **"@AmericanAir Thank you, you too!"** was labelled Neutral but predicted as Positive with 97.5% confidence because of the strong *"thank you"* cue. This may also reflect some ambiguity in the original dataset labels.

Another example, **"@AmericanAir I wish I could remember all of their names!"**, was labelled Positive but predicted as Neutral with only 50.1% confidence. Here, the positive sentiment is expressed more through tone than through obvious positive words, making it harder for the model to detect. These examples show why **Neutral sentiment remains the most challenging class** for both models.

Business Recommendations

- Based on the model results and the deployed dashboard, **Logistic Regression is recommended as the default model for regular monitoring** because it is faster, smaller, and easier to interpret. The **Bidirectional LSTM** can be used when higher accuracy is more important and the additional training and computational cost is acceptable.
- Tweets with **low-confidence predictions should be reviewed by a human**, especially because many uncertain cases involve the difficult Neutral class. A sudden increase in negative tweets containing words such as *"delayed", "cancelled",* or *"luggage"* can also be treated as an early warning of possible operational issues. At the same time, Neutral predictions should not automatically be considered as "no action needed", since both models have shown weaker performance on this class.

Conclusion

This project developed a complete NLP pipeline for Twitter sentiment analysis, covering **data preprocessing, exploratory analysis, feature engineering, model training, evaluation, error analysis, interpretability, and deployment** through the BrandPulse AI dashboard.

The **Bidirectional LSTM achieved the best performance**, with **80.0% accuracy and 0.80 weighted F1**, compared with **77.4% accuracy and 0.78 weighted F1** for TF-IDF with Logistic Regression. However, Logistic Regression remains a practical choice because it requires much less training time, has a much smaller model size, and provides clearer word-level explanations.

Both approaches struggled with the **Neutral** class, which had an F1-score of around **0.60**. This was mainly due to tweets with weak or unclear sentiment signals. The final BrandPulse AI dashboard brings the complete solution together with **single-tweet prediction, batch CSV analysis, sentiment visualizations, simulated live-tweet monitoring, and model comparison**, making the system practical for exploring and monitoring customer sentiment.

Future Scope

- The project can be improved further in several ways. More **Neutral-class examples** can be collected and labelled to help address the main weakness identified in the current models. A **transformer-based model such as DistilBERT** can also be evaluated as a third approach to capture richer contextual information. Other improvements include adding **sarcasm detection**, replacing the simulated live stream with a real Twitter/X API feed, and expanding the dataset beyond the airline domain so the system can be applied to other industries.
- For deployment, the trained models can also be exposed through a **lightweight REST API**, allowing other applications or internal systems to use the sentiment predictions directly in addition to the Streamlit dashboard.